



Ordering numbers in standard form

Task A: Comparing numbers in standard form

1) Put the following in ascending order, smallest to largest.

a) 3×10^{12} 1×10^{13} 4×10^{11} 2×10^{10} 9×10^9

b) 2.3×10^4 2.3×10^{-1} 2.3×10^{-4} 2.3×10^{-3} 2.3×10

c) 4.99×10^6 4×10^6 9.8×10^{-7} 4.99×10^{-6} 4.9×10^{-6}

d) 9.7×10^{-2} 1.001×10^9 9.89×10^{-2} 1.5×10^{-4} 1.01×10^9

2) Put the following in ascending order in the format they are given.

a) 1×10^3 0.0089 0.1 8.2×10^{-4} 90 000

b) 21 000 0.000 087 0.0002 2.01×10^4 0.000 078

3) Laura says



Laura

I have put the following in ascending order by ordering the exponents first and where the exponents are the same, I have ordered the number.

Explain why Laura has not correctly ordered the numbers.

0.6×10^{-1}

6×10^2

6.2×10^2

0.618×10^3

Task B: Ordering numbers using standard form

1) Using inequalities, $<$, $>$ or $=$, compare the sizes of each number, making sure to show your working out.

a) 38×0.01 4.8×10^{-2}

b) $78\,000 \times 0.001$ 99×10^2

c) $139 \times \frac{1}{10^2}$ $4820 \times \frac{1}{10^4}$

d) $1600 \times \frac{1}{10^2}$ $160\,000 \times \frac{1}{10^4}$

2) By working out the number, write in standard form and order the original form given in the question.

a) 3.8×0.001 7900×10^{-4} $7.8 \times \frac{1}{10^2}$ $800 \times \frac{1}{10^{-2}}$

b) 920×0.0001 $92\,100 \times 10^{-3}$ $9.8 \times \frac{1}{10}$ $9100 \times \frac{1}{10^{-2}}$

c) 480×0.1 4.8×10^{-2} $4.81 \times \frac{1}{10^2}$ $4820 \times \frac{1}{10^4}$

